

BROWNWOOD, TX, USA

WMO: 722666

Lat: 31.800N

Lon: 98.950W

Elev: 420

StdP: 96.38

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 93943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-4.0	-13.7	1.2	-0.2	-11.3	1.5	3.2	11.6	11.1	10.6	13.6	1.9	0	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	38.0	22.1	37.4	22.2	36.1	22.5	25.5	32.4	24.9	31.8	24.3	31.0	4.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.6	27.4	22.9	18.6	26.9	22.6	18.2	26.7	80.5	32.7	78.2	32.2	75.9	30.0	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.6	7.7	DB -9.9	39.8	1.9	1.7	-11.3	41.0	-12.4	42.0	-13.5	42.9	-14.9	44.1		
(5)			WB -10.7	26.2	1.9	1.1	-12.0	27.0	-13.1	27.6	-14.2	28.2	-15.5	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	7.9	9.8	14.7	18.5	22.8	27.0	28.5	28.6	24.8	19.1	13.0	8.4		
	DBStd	8.61	4.98	5.88	5.14	4.42	4.04	2.42	1.96	2.40	3.11	4.76	5.31	5.10		
	HDD10.0	313	99	68	17	2	0	0	0	0	0	3	31	92		
	HDD18.3	1288	325	243	132	50	11	0	0	0	1	48	171	309		
	CDD10.0	3468	34	62	164	258	396	511	573	576	444	286	121	43		
	CDD18.3	1401	1	4	21	56	148	261	314	318	194	72	11	1		
	CDH23.3	16877	26	83	248	678	1571	3094	3995	4086	2126	791	142	36		
	CDH26.7	8634	3	19	65	249	704	1619	2260	2362	1033	283	31	5		
(14) Wind	WSAvg	3.3	3.1	3.6	3.9	4.2	3.9	3.7	3.1	2.8	2.7	2.9	3.2	3.0		
(15) Precipitation	PrecAvg	811	35	52	57	66	96	104	54	58	65	73	45	34		
	PrecMax	1149	98	185	173	255	263	306	164	248	137	164	163	204		
	PrecMin	521	0	2	1	6	30	16	1	0	5	3	0	0		
	PrecStd	179	25	49	42	54	54	69	45	66	43	55	48	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	29.8	31.1	35.3	37.4	38.7	39.8	40.0	37.5	33.8	30.0	27.3		
		MCWB	14.6	16.2	16.9	18.0	22.1	21.6	21.2	22.4	21.0	20.9	18.8	15.5		
	2%	DB	23.0	26.0	28.7	32.2	34.9	37.2	37.8	38.0	35.1	32.2	27.3	23.7		
		MCWB	14.2	14.9	16.9	18.3	22.1	22.2	22.2	22.3	21.5	21.3	17.8	14.5		
	5%	DB	20.9	22.9	26.3	29.0	32.7	35.2	37.2	37.4	33.7	29.8	24.8	21.1		
		MCWB	13.6	14.2	16.6	18.1	21.6	22.4	22.5	22.4	21.8	20.0	16.5	13.5		
	10%	DB	17.8	20.1	23.8	27.4	30.3	33.8	35.2	36.1	32.4	27.8	22.5	17.9		
		MCWB	11.5	13.4	15.7	18.2	21.1	22.8	22.5	22.9	21.9	18.8	15.9	12.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	19.1	20.9	22.8	25.3	26.6	25.8	25.9	25.3	24.2	21.2	19.3		
		MCDB	22.0	23.3	26.4	28.4	31.5	32.2	32.7	33.6	32.4	30.4	26.3	22.6		
	2%	WB	17.0	17.8	19.5	21.6	24.2	25.5	25.1	25.2	24.3	22.9	19.9	17.1		
		MCDB	19.9	21.7	24.3	26.9	30.1	31.8	32.4	32.9	30.4	27.7	23.7	20.5		
	5%	WB	14.5	16.3	18.6	20.6	23.4	24.6	24.5	24.6	23.7	22.1	18.6	15.1		
		MCDB	18.5	20.6	23.0	26.0	29.3	30.8	31.7	31.9	29.5	26.5	22.3	18.6		
	10%	WB	12.2	14.5	17.6	19.8	22.6	23.9	24.0	24.0	23.0	21.2	17.4	13.1		
		MCDB	17.0	19.2	21.4	24.3	28.4	30.0	31.0	31.1	28.5	25.4	20.7	17.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	14.6	14.3	14.6	12.7	12.3	13.0	13.3	13.0	14.1	14.5	15.0		
		MCDBR	19.8	19.6	17.4	18.1	16.1	14.6	15.2	15.2	15.1	15.6	17.5	19.9		
		MCWBR	11.2	10.7	8.2	8.3	5.9	4.1	3.9	3.8	4.9	6.7	9.4	11.6		
	5% WB	MCDBR	15.9	15.8	12.3	13.4	12.3	11.5	12.5	12.9	11.9	11.6	12.8	14.3		
		MCWBR	10.7	10.2	6.9	7.4	5.4	4.6	4.1	4.1	4.5	5.8	8.3	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.310	0.335	0.376	0.413	0.414	0.425	0.417	0.389	0.353	0.321	0.305		
	taud		2.521	2.505	2.449	2.357	2.313	2.347	2.366	2.371	2.393	2.481	2.500	2.519		
	Ebn at Noon		929	953	947	914	877	872	860	865	879	892	897	905		
	Edh at Noon		85	95	108	124	131	126	123	121	114	97	86	81		
(44) All-Sky Solar Radiation	RadAvg		3.14	3.78	4.88	5.89	6.16	6.91	6.87	6.30	5.29	4.23	3.30	2.79		
	RadStd		0.33	0.54	0.40	0.50	0.61	0.55	0.50	0.43	0.43	0.64	0.42	0.36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	N/A

Nomenclature: See separate page